

# Machine Learning Based Forecasting of Dengue Fever Using Environmental Time Series Data

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## Abstract

Accurate forecasting of dengue fever cases is essential for effective public health planning and early intervention. In this study, we address the dengue case prediction problem as part of the DengAI competition hosted by DrivenData, where weekly dengue case counts are predicted using environmental and climatic variables. We propose and evaluate multiple predictive models, including a polynomial regression baseline and a Gradient Boosting Regression model both trained separately for cities San Juan and Iquitos to consider location-specific outbreak dynamics. Model performance is assessed using Mean Absolute Error (MAE) under a time-based evaluation protocol.

In addition to weekly prediction, we further investigate early prediction by forecasting dengue cases up to four weeks in advance. Our results show that Gradient Boosting Regression substantially outperforms polynomial regression for both cities, and that prediction accuracy generally decreases as the forecasting horizon increases. The findings highlight the importance of nonlinear ensemble models and short-term forecasting for effective dengue outbreak prediction.

## CCS Concepts

• **Computing methodologies** → **Machine learning**; *Ensemble methods*; • **Mathematics of computing** → *Time series analysis*; • **Applied computing** → Health informatics.

## Keywords

Dengue fever prediction, machine learning, time series forecasting, gradient boosting, ensemble learning, early prediction, regression models

## 1 Introduction

Dengue fever is a mosquito-borne viral disease that poses a major public health challenge in tropical and subtropical regions. Accurate forecasting of dengue outbreaks can support early warning systems, optimize resource allocation, and enable preventive interventions. However, forecasting dengue incidence remains a challenging task due to complex interactions between climatic, environmental, and temporal factors.

In this study, we participate in the DengAI competition hosted by DrivenData, which focuses on predicting weekly dengue case counts using environmental and climatic variables [2]. We begin with a polynomial regression model as a nonlinear baseline and examine the effect of polynomial degree on predictive performance. We then apply Gradient Boosting Regression, an ensemble-based approach capable of modeling nonlinear relationships and feature interactions without explicit feature engineering. To ensure fair

evaluation, all models are trained and validated using a time-based split that respects the temporal ordering of observations.

In addition to standard weekly prediction, we address the problem of early prediction by forecasting dengue cases several weeks ahead. This setting is particularly relevant for public health applications, where advance warning can improve preparations. We evaluate forecasting horizons ranging from one to four weeks and identify the most reliable prediction window under the current modeling framework.

## 2 Related Work

Dengue fever forecasting has been studied using statistical and machine learning approaches, focusing on environmental and climatic drivers. Recent studies have explored the use of machine learning techniques for predicting dengue outbreaks using environmental and climatic factors. Rahman et al. [4] propose an artificial intelligence framework to predict dengue outbreaks based on eco-climatic triggers, highlighting the importance of humidity, temperature, and seasonal patterns. Madewell et al. [3] apply machine learning models to predict severe dengue cases in Puerto Rico, demonstrating the effectiveness of data-driven approaches in public health forecasting.

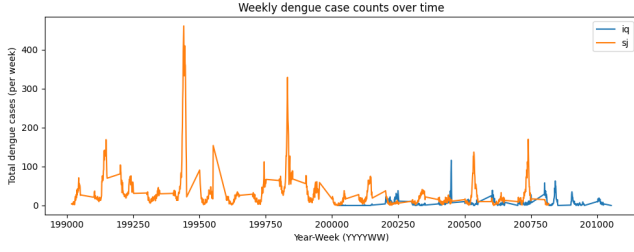
Tree-based ensemble models, including Gradient Boosting, also have been frequently applied to the DengAI dataset and similar epidemiological forecasting tasks due to their ability to model nonlinear feature interactions without extensive feature engineering. Several public implementations and competition-based studies report strong performance using ensemble methods for short-term dengue prediction [1].

## 3 Dataset

The dataset used in the study is provided by the DengAI competition hosted by DrivenData [2]. It contains weekly dengue case records along with environmental variables for two cities: San Juan and Iquitos. Features include daily weather and precipitation information with reanalysis measurements. The target variable is total number of reported dengue fever cases for each week.

Figure 1 illustrates the weekly dengue case counts over time for San Juan and Iquitos. San Juan shows substantially higher and more frequent peaks, with weekly case counts reaching over 400 in certain periods, whereas Iquitos generally exhibits lower case counts with extended periods of minimal activity. These differences indicate distinctive outbreak dynamics between the two locations.

Table 1 provides a summary of the dataset statistics for each city. While both cities share the same number of input features, San Juan contains a substantially larger number of weekly disease count in addition to higher proportion of missing feature values. The target variable distributions also differ between cities, with



**Figure 1: Weekly dengue case counts over time for San Juan (sj) and Iquitos (iq). San Juan exhibits higher outbreak intensity and more frequent peaks, while Iquitos shows lower and more discontinuous case counts.**

San Juan having a higher median and maximum number of cases. These characteristics highlight the need for careful preprocessing and city-aware modeling strategies.

**Table 1: Summary statistics of the DengAI training dataset for each city, including the number of weekly observations, number of features, average feature missingness, and target variable range.**

| City          | Weeks | Feat. | Avg. miss. | Cases (min/med/max) |
|---------------|-------|-------|------------|---------------------|
| Iquitos (iq)  | 520   | 21    | 0.015      | 0 / 5 / 116         |
| San Juan (sj) | 936   | 21    | 0.193      | 0 / 19 / 461        |

### 3.1 Data Preprocessing

**3.1.1 Missing Value Handling.** The dataset contains missing values in several environmental variables, with missing rates varying between the two cities. To protect city feature distributions, we performed processing separately for each city. For each numeric feature, missing entries are replaced with the mean values of known observations of the same city.

**3.1.2 City-Base Grouping.** Due to San Juan and Iquitos showing different dengue case dynamics and different level of missing value rates, we processed and later modeled each city separately. For each city, data is divided into two sets and sorted chronologically using time features. Then for each city grouped sets, 80% of the weeks were taken for training while remaining %20 were used for validation. In the end, for San Juan we had 748 train, 188 validation samples. For Iquitos we had 416 train, 104 validation samples.

## 4 Methodology

This section describes the predictive models developed in this study. We evaluate two main classes of models. The first is a high-degree polynomial regression model, where the degree is selected based on validation performance. The second class consists of gradient boosting regression model inspired by prior literature on dengue forecasting[1].

### 4.1 Baseline and Polynomial Regression Model

As a baseline nonlinear regression approach, we employ polynomial regression to model the relationship between environmental variables and weekly dengue case counts. For each city, we construct a design matrix by expanding the original input features with polynomial terms up to degree  $k$  using `PolynomialFeatures` (without an explicit bias term), and then fit an ordinary least squares linear model (`LinearRegression`). Model performance is evaluated using Mean Absolute Error (MAE) on the time-based validation split.

Because the DengAI dataset contains two cities with distinct outbreak dynamics, we train separate polynomial regression models for San Juan and Iquitos. We initially set the polynomial degree to  $k = 2$ , and then select the degree from  $\{2, 3, 4\}$  based on validation MAE for each city. This procedure provides a simple baseline for capturing nonlinear relationships and serves as a reference point for more advanced machine learning models.

### 4.2 Gradient Boosting Regression

As a second approach inspired by previous dengue case forecasting studies, we used Gradient Boosting Regression (GBR)[1]. Gradient boosting builds an additive ensemble of shallow regression trees, where each successive tree is trained to reduce the residual errors of the current model. This approach can capture nonlinear relationships and feature interactions.

We train separate GBR models for San Juan and Iquitos using the same time-based split described in the preprocessing section. The model is configured with 300 estimators, learning rate of 0.05, and a maximum tree depth of 3. The selected hyperparameters were chosen to balance model capacity and generalization, using shallow trees (depth=3) and a small learning rate to reduce overfitting while allowing sufficient ensemble complexity. Performance is evaluated using Mean Absolute Error(MAE) on the validation period for each city.

To identify which environmental variables contribute most to predictive performance, we analyze feature importance scores provided by the GBR models. These scores reflect the average reduction in the training loss attributable to splits using each feature across the boosted trees. In results section We report the top-ranked features separately for each city to highlight potential differences in outbreak drivers between locations.

### 4.3 Forecasting Horizon Evaluation

In addition to weekly predictions, we evaluate models by forecasting dengue cases  $h$  weeks in advance. For a given horizon  $h$ , we construct a shifted target variable  $y_{t+h}$  by moving the original weekly case counts forward by  $h$  weeks, while keeping the input feature vector at time  $t$  unchanged. For each city we define  $\text{future\_cases}_t = \text{total\_cases}_{t+h}$  and discard the last  $h$  observations for which the future target is unavailable.

We train and evaluate a Gradient Boosting Regression model for each horizon  $h \in \{1, 2, 3, 4\}$  using the same chronological 80/20 split as in the weekly prediction setting. The best early prediction period is defined as the horizon which produces lowest MAE score.

## 5 Experimental Setup

All experiments are done using the DengAI dataset with a time-based evaluation protocol. We avoid random shuffling to ensure that training data always precedes validation data chronologically. For each city, data is separated into training and validation sets with 80/20 ratio.

Model performance is evaluated using Mean Absolute Error (MAE), which is the official evaluation metric of the DengAI competition. MAE measures the average absolute difference between predicted and true weekly dengue case counts and provides an interpretable error measure in the same units as the target variable.

We compare two classes of models: polynomial regression models with varying polynomial degrees, and gradient boosting regression models. All models are trained and evaluated separately for San Juan and Iquitos using the same preprocessing pipeline, feature set, and evaluation protocol to ensure fair comparison.

To quantify early warning capability, we evaluate forecasting horizons of 1 to 4 weeks ahead. For each horizon, we retrain the model using the horizon-shifted target and compute MAE on the held-out validation period. Results are reported separately for each city, and we select the best horizon based on minimum validation MAE.

## 6 Results

### 6.1 Weekly Prediction Models

Results of the first section of the project where we develop and compare multiple machine learning models, to determine the best option to predict dengue case counts on a weekly basis.

Table 2 reports validation MAE for polynomial regression with degrees  $k \in \{2, 3, 4\}$ . For both cities, performance decreases as the polynomial degree increases, with particularly severe decline for San Juan. These results indicate strong overfitting effects when high-degree polynomial expansions are used without regularization. As a result, we select  $k = 2$  as the best polynomial baseline. Comparing the MAE differences between each city, we can guess that Iquitos having lower scores, which indicates easier predictability, is due to San Juan having higher variance and complexity.

**Table 2: Validation MAE of polynomial regression for different polynomial degrees  $k$  using weekly prediction.**

| City     | $k = 2$ | $k = 3$ | $k = 4$ |
|----------|---------|---------|---------|
| San Juan | 53.52   | 303.77  | 646.93  |
| Iquitos  | 15.41   | 46.05   | 55.96   |

Table 3 compares weekly prediction performance between the polynomial baseline and Gradient Boosting Regression. The gradient boosting model substantially outperforms polynomial regression for both cities, reducing MAE by a large amount. This improvement highlights the advantage of ensemble-based tree models to capture nonlinear relationships and feature interactions present in the dengue dataset, especially in San Juan where peaks cases are more volatile.

**Table 3: Comparison of weekly prediction performance (MAE) between polynomial regression ( $k = 2$ ) and Gradient Boosting Regression.**

| City          | Polynomial ( $k = 2$ ) | Gradient Boosting |
|---------------|------------------------|-------------------|
| San Juan (sj) | 53.52                  | <b>18.31</b>      |
| Iquitos (iq)  | 15.41                  | <b>7.51</b>       |

### 6.2 Feature Importance Analysis

Feature importance analysis from the Gradient Boosting models reveals that temporal variables such as year and weekofyear are among the most influential predictors for both cities, reflecting strong seasonal patterns in dengue cases. In addition, humidity- and temperature-related features consistently rank among the top contributors, aligning with known environmental drivers of dengue transmission. Differences in feature rankings between San Juan and Iquitos further suggest location specific outbreak dynamics. These differences justify the use of city specific models.

**Table 4: Top 10 features ranked by Gradient Boosting importance for San Juan.**

| Feature                               | Importance |
|---------------------------------------|------------|
| year                                  | 0.524      |
| weekofyear                            | 0.175      |
| reanalysis_specific_humidity_g_per_kg | 0.068      |
| reanalysis_dew_point_temp_k           | 0.047      |
| station_max_temp_c                    | 0.030      |
| ndvi_se                               | 0.024      |
| reanalysis_max_air_temp_k             | 0.020      |
| station_precip_mm                     | 0.019      |
| reanalysis_avg_temp_k                 | 0.016      |
| reanalysis_precip_amt_kg_per_m2       | 0.013      |

**Table 5: Top 10 features ranked by Gradient Boosting importance for Iquitos.**

| Feature                               | Importance |
|---------------------------------------|------------|
| year                                  | 0.240      |
| weekofyear                            | 0.190      |
| reanalysis_specific_humidity_g_per_kg | 0.102      |
| reanalysis_avg_temp_k                 | 0.060      |
| reanalysis_tdtr_k                     | 0.056      |
| ndvi_sw                               | 0.054      |
| reanalysis_dew_point_temp_k           | 0.047      |
| reanalysis_relative_humidity_percent  | 0.037      |
| reanalysis_precip_amt_kg_per_m2       | 0.032      |
| ndvi_se                               | 0.030      |

### 6.3 Forecasting Horizon Evaluation

To assess early warning capability, we train models to predict dengue case counts several weeks ahead of time. We evaluate how

prediction accuracy degrades as the forecasting horizon increases and identify the optimal lead time for early prediction.

For Iquitos, while the 1-week horizon yields the best overall performance, the 3-week horizon remains close, whereas the 4-week horizon exhibits a sudden decrease in accuracy. Overall, these results suggest that a 1-week-ahead forecast provides the most reliable early prediction for both cities under the current model and feature set.

Table 6 summarizes validation MAE for early prediction using Gradient Boosting Regression at forecasting horizons ranging from 1 to 4 weeks. For San Juan, prediction error increases consistently as the forecast horizon grows, indicating a gradual loss of predictive accuracy when forecasting further into the future. The lowest MAE is achieved at a 1-week horizon.

**Table 6: Validation MAE of Gradient Boosting Regression for early prediction at different forecasting horizons  $h$  (in weeks). Lower values indicate better performance.**

| City          | $h = 1$ | $h = 2$ | $h = 3$ | $h = 4$ |
|---------------|---------|---------|---------|---------|
| San Juan (sj) | 17.54   | 17.82   | 18.25   | 18.72   |
| Iquitos (iq)  | 7.57    | 9.15    | 8.06    | 11.17   |

The final best performing model (Gradient Boosting) is trained on the complete training data and used to generate predictions for the DengAI test set, which are submitted to the competition platform. As final result we got 25.4014 MSE scoring.

## 7 Conclusion

In this study, we investigated machine learning approaches for predicting weekly dengue fever case counts using environmental and climatic data from the DengAI competition. We compared a polynomial regression baseline with a Gradient Boosting Regression model and demonstrated that ensemble based methods substantially outperform polynomial models for both San Juan and Iquitos.

We further evaluated early prediction by forecasting dengue cases up to four weeks in advance. Our results show that prediction accuracy generally degrades as the forecasting horizon increases, with the most reliable performance achieved at a 1-week-ahead forecast for both cities. This finding suggests that short-term early warning systems are more effective under the current feature set and modeling framework.

Overall, the results highlight the importance of nonlinear modeling and careful handling of temporal structure in dengue forecasting tasks. Future work may explore models with explicit temporal memory, such as auto regressive or sequence-based learning approaches, to improve longer-horizon prediction performance and enhance early warning capabilities.

## References

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